

修士論文

Microlocal Morse Theory and Truncated
Microsupport
(超局所モース理論と打ち切り超局所台)

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1 Introduction

In this thesis, We study microlocal Morse theory, which is presented, for instance, in [GKS12]. However, we replace the notion of microsupports with that of truncated microsupprts, introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KFS03]. In [KFS03], they do not study the functorial property of truncated microsupport of direct images of non proper morphisms. We prove this functorial property under non proper direct images.

2 Notations and Reviews

We consider a commutative unital ring $\mathbf{k}$ of finite global dimension (e.g. $\mathbf{Z}$ or a field). Let X be a topological space. We denote by $D^b(\mathbf{k}_X)$ the bounded derived category of sheaves of $\mathbf{k}$ -modules on X .

Let us denote Z a locally closed set of X and $i: Z \hookrightarrow X$ an inclusion. We use the following functors:

$$R\Gamma_Z(\cdot) := Ri_*i^!, \quad (\cdot)_Z := Ri_!i^{-1}.$$

If there is no risk, we will write $\mathbf{k}_Z$ rather than $\mathbf{k}_{X,Z}$.

Geometrical Notations

Let M be a real manifold of class C^∞ . We denote by $\pi: T^*M \rightarrow M$ the cotangent bundle of M . If $f: M \rightarrow N$ is a morphism of manifolds, then we have the natural diagram.

$$\begin{array}{ccccc} TM & \xrightarrow{f'} & M \times_N TN & \xrightarrow{f_\tau} & TN \\ & \searrow \tau_M & \downarrow N & \square & \downarrow \tau_N \\ & & M & \xrightarrow{f} & N, \end{array} \quad \begin{array}{ccccc} T^*M & \xleftarrow{f_d} & M \times_N T^*N & \xrightarrow{f_\pi} & T^*N \\ & \searrow \pi_M & \downarrow N & \square & \downarrow \pi_N \\ & & M & \xrightarrow{f} & N. \end{array}$$

In this diagram, f_τ and f_π denote the natural projections. f' is defined by $(x; v) \mapsto (x; df_x v)$, and f_d is induced by the transposition of f' .

3 Truncated Microsupport

Definition 3.1. Let $F \in D^b(\mathbf{k}_X)$. The closed conic subset $SS_k(F)$ of T^*X is defined by: $p \notin SS_k(F)$ if and only if there exists an open conic neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1 -function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

$$(3.1) \quad H_{\{\varphi \geq 0\}}^j(F)_x = 0 \quad \text{for any } j \leq k.$$

Functorial Properties

Proposition 3.2. Let $f: X \rightarrow Y$ be a morphism of real analytic manifolds and let $F \in D^b(\mathbf{k}_X)$ such that f is proper on $\text{supp}(F)$. Then for any $k \in \mathbf{Z}$,

$$(3.2) \quad SS_k(Rf_* F) \subset f_\pi f_d^{-1}(SS_k(F)).$$

The equality holds in case f is a closed embedding.

4 Lemmas from [KS90]

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary.

For a projective system of abelian groups $(M_n)_n$, we set

$$M_\infty := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$(4.1) \quad E_n^\bullet: \cdots \rightarrow M_n^{j-1} \rightarrow M_n^j \rightarrow M_n^{j+1} \rightarrow \cdots,$$

and projective limit

$$(4.2) \quad E_\infty^\bullet: \cdots \rightarrow M_\infty^{j-1} \rightarrow M_\infty^j \rightarrow M_\infty^{j+1} \rightarrow \cdots.$$

Denote by

$$(4.3) \quad \Phi_k: H^k(E_\infty^\bullet) \rightarrow \varprojlim_n H^k(E_n^\bullet)$$

the natural morphism.

Proposition 4.1 ([KS90, Proposition 1.12.4]). Assume that for each $j \in \mathbf{Z}$, the system $(M_n^j)_n$ satisfies the ML condition. Then

- (a) for each $k \in \mathbf{Z}$, the map Φ_k is surjective,
- (b) if moreover, for a given i the system $(H^{i-1}(E_n^\bullet))_n$ satisfies the ML condition, then Φ_i is bijective.

The following is called Kashiwara's lemma.

Proposition 4.2 ([KS90, Proposition 1.12.6]). Let $(X_s, \rho_{s,t}: X_t \rightarrow X_s)_{(s, (s \leq t))}$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps

$$\lambda_s: X_s \rightarrow \varprojlim_{r < s} X_r$$

and

$$\mu_s: \varinjlim_{t > s} X_t \rightarrow X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0, s_1} ($s_0 \leq s_1$) are injective (resp. surjective).

ML procedure for sheaves

Proposition 4.3 ([KS90, Proposition 2.7.1]). Let X be a topological space and let $F \in \mathbf{D}^+(\mathbf{Z}_X)$. Let $(U_n)_{n \in \mathbf{N}}$ be an increasing sequence of open subsets of X , $(Z_n)_{n \in \mathbf{N}}$ a decreasing sequence of closed subsets of X . Set $U := \bigcup_n U_n$, $Z := \bigcap_n Z_n$.

- (i) For any j , the natural map $\varphi_j: H_Z^j(U; F) \rightarrow \varprojlim_n H_{Z_n}^j(U_n; F)$ is surjective.
- (ii) Assume that for a given j , the projective system $(H_{Z_n}^{j-1}(U_n; F))_n$ satisfies the ML condition. Then φ_j is bijective.
- (iii) Let now $(X_n)_{n \in \mathbf{N}}$ be an increasing family of subsets of X satisfying $X = \bigcup_n X_n$ and $X_n \subset \text{Int}(X_{n+1})$ for all n . Assume that for a given j , the projective system $(H^{j-1}(X_n; F))_n$ satisfies the ML condition. Then the natural map $H^j(X; F) \rightarrow \varprojlim_n H^j(X_n; F)$ is bijective.

5 Truncated Noncharacteristic Deformation

We will reformulate the noncharacteristic deformation lemma [KS90, Proposition 2.7.2] to make a relation to truncated microsupport.

Proposition 5.1. Let X be a locally compact (i.e., by definition, Hausdorff) space countable at infinity (i.e., countable union of compact subspaces). Let $F \in D^b(\mathbf{Z}_X)$, and $(U_t)_{t \in \mathbf{R}}$ be a family of open subsets of X . Let $k \in \mathbf{Z}$. If s is a real number, set $Z_s := \bigcap_{t > s} \overline{U_t - U_s}$. We assume the following conditions.

- (i) $U_t = \bigcup_{s < t} U_s$, for all $t \in \mathbf{R}$.
- (ii) For all pairs (s, t) with $s \leq t$, the set $\overline{U_t - U_s} \cap \text{supp } F$ is compact.
- (iii) For all pairs (s, t) with $s \leq t$, and all $x \in Z_s - U_t$, we have

$$(H_{X-U_t}^j(F))_x = 0 \quad \text{for all } j \leq k.$$

- (iv) For any $s \in \mathbf{R}$, $Z_s \subset U_t$.

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$H^j \left(\bigcup_{s \in \mathbf{R}} U_s; F \right) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for all } j < k,$$

and the injections

$$H^k \left(\bigcup_{s \in \mathbf{R}} U_s; F \right) \hookrightarrow H^k(U_t; F).$$

Remark 5.2. Let us give an example that satisfies the hypotheses (i)–(iii) but not (iv). Let us define a locally closed subset Z of $\mathbf{R}^2$ by

$$Z := \{(x, y) \in \mathbf{R}^2 ; (x+2)^2 + (y-1)^2 = 1\} \setminus \{(-2, 0)\}.$$

Consider a real valued function φ_s for a real number s :

$$(5.1) \quad \varphi_s(x) := \begin{cases} 0 & |x| \geq 2, \\ \max(0, s(2 - |x|)) & |x| \leq 2. \end{cases}$$

We define $(U_t)_{t \in \mathbf{R}}$, a family of open sets of $\mathbf{R}^2$ by

$$U_t := \{(x, y) \in \mathbf{R}^2 ; y < \varphi_s(x)\}.$$

Then $F := \mathbf{k}_Z$ and $(U_t)_t$ satisfy the hypotheses for $k = 0$. Therefore, the natural morphism

$$H^0(U_1; \mathbf{k}_Z) = \mathbf{k} \rightarrow H^0(U_0; \mathbf{k}_Z) = 0$$

is not injective and the consequence of Proposition 5.1 does not hold.

Proof of Proposition 5.1. Consider the following conditions for $j \in \mathbf{Z}$, s , and t :

$$\begin{aligned} (a)_s^j & \quad \mu_s^j : \varinjlim_{t > s} H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F), \\ (a')_s^k & \quad \mu_s^k : \varinjlim_{t > s} H^k(U_t; F) \hookrightarrow H^k(U_s; F), \\ (b)_s^j & \quad \lambda_s^j : H^j(U_t; F) \xrightarrow{\sim} \varprojlim_{s < t} H^j(U_s; F). \end{aligned}$$

Let us prove $(a)_s^j$ for $s \in \mathbf{R}$ and $j < k$. We can assume that $\overline{U_t - U_s}$ is compact for any $s \leq t$ by replacing X by $\text{supp } F$. Consider the distinguished triangle

$$\text{R}\Gamma_{X-U_t}(F) \rightarrow F \rightarrow \text{R}\Gamma_{U_t}(F) \xrightarrow{+1}.$$

Applying to this triangle the functor $(\cdot)_{\overline{U_t}}$, we have

$$(\text{R}\Gamma_{X-U_t}(F))_{\overline{U_t}} \rightarrow F_{\overline{U_t}} \rightarrow \text{R}\Gamma_{U_t}(F) \xrightarrow{+1}.$$

Again applying to this triangle the functor $\text{R}\Gamma(X; \cdot)$, we have

$$(5.2) \quad \text{R}\Gamma(X; (\text{R}\Gamma_{X-U_t}(F))_{\overline{U_t}}) \rightarrow \text{R}\Gamma(\overline{U_t}; F) \rightarrow \text{R}\Gamma(U_t; F) \xrightarrow{+1}.$$

By hypotheses (iii)–(??), it follows that

$$\left(H_{X-U_t}^j(F) \right)_{\partial U_t} = 0$$

for any $j \leq k$ and then

$$H^j(X; (\text{R}\Gamma_{X-U_t}(F))_{\overline{U_t}}) = 0 \quad \text{for any } j \leq k.$$

*1 Thus for any t , we have

$$(5.3) \quad H^j(\overline{U_t}; F) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for any } j < k,$$

$$(5.4) \quad H^k(\overline{U_t}; F) \hookrightarrow H^k(U_t; F).$$

Therefore, we have by (iv) and (5.3),

$$\begin{aligned} \varinjlim_{t>s} H^j(U_t; F) &\cong H^j\left(\bigcap_{t>s} U_t; F\right) \\ &= H^j(\overline{U_s}; F) \\ &\xrightarrow{\sim} H^j(U_s; F) \end{aligned}$$

for any $j < k$, and $(a)_s^j$ follows. Similarly, we have by (iv) and (5.4),

$$\begin{aligned} \varinjlim_{t>s} H^k(U_t; F) &\cong H^k\left(\bigcap_{t>s} U_t; F\right) \\ &= H^k(\overline{U_s}; F) \\ &\hookrightarrow H^k(U_s; F) \end{aligned}$$

and $(a')_s^k$ follows. □

*1 この辺の議論はきちんと結めないといけない。

6 Relative Microlocal Morse Theory via Truncated Microsupport

Let $f: Y \rightarrow X$ be a morphism of manifolds, and φ a real valued function of class C^1 . For a real number $t \in \mathbf{R}$, we set

$$\begin{aligned} Y_t &:= \{y \in Y ; \varphi(y) < t\} = \varphi^{-1}((-\infty, t)), \\ \tilde{Y}_t &:= \{y \in Y ; \varphi(y) \leq t\} = \varphi^{-1}((-\infty, t]). \end{aligned}$$

Let us define the following morphisms:

$$\begin{array}{ccc} Y_t & \xrightarrow{j_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X, \end{array} \quad \begin{array}{ccc} \tilde{Y}_t & \xrightarrow{\tilde{j}_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X. \end{array}$$

Proposition 6.1. Let $G \in D^b(Y)$, and assume $f_t: \text{supp}(G) \cap \bar{Y}_t \rightarrow X$ is proper for all $t \in \mathbf{R}$. Let k be an integer and t_o a real number.

(i) Assume that, for any $y \in Y - \tilde{Y}_{t_o}$, we have

$$d\varphi(y) \notin \left(\text{SS}_k(G) + f_d \left(Y \times_X T^*X \right) \right).$$

Then

$$\begin{aligned} \text{(a)} \quad & R^j f_* G \cong R^j f_{t*} j_t^{-1} G \quad \text{for } j < k \text{ and } t > t_o, \\ \text{(a}_k\text{)} \quad & R^k f_* G \hookrightarrow R^k f_{t*} j_t^{-1} G \quad \text{for } t > t_o, \\ \text{(a')} \quad & R^j \tilde{f}_* G \cong R^j \tilde{f}_{t*} \tilde{j}_t^{-1} G \quad \text{for } j < k \text{ and } t \geq t_o, \\ \text{(a}'_k\text{)} \quad & R^k \tilde{f}_* G \hookrightarrow R^k \tilde{f}_{t*} \tilde{j}_t^{-1} G \quad \text{for } t \geq t_o, \\ \text{(b)} \quad & \text{SS}_k(Rf_* G) \subset f_\pi \left(f_d^{-1} \left(\text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\ & = f_\pi f_d^{-1} (\text{SS}_k(G)). \end{aligned}$$

(ii) Assume that, for any $y \in Y - \tilde{Y}_{t_o}$, we have

$$-d\varphi(y) \notin \left(\text{SS}_k(G) + f_d \left(Y \times_X T^*X \right) \right).$$

Then

$$\begin{aligned} \text{(c)} \quad & R^j f_! G \cong R^j f_{t!} j_t^{-1} G \quad \text{for } j < k \text{ and } t > t_o, \\ \text{(c}_k\text{)} \quad & R^k f_! G \hookrightarrow R^k f_{t!} j_t^{-1} G \quad \text{for } t > t_o, \\ \text{(c')} \quad & R^j \tilde{f}_! G \cong R^j \tilde{f}_{t!} \tilde{j}_t^{-1} G \quad \text{for } j < k \text{ and } t \geq t_o, \\ \text{(c}'_k\text{)} \quad & R^k \tilde{f}_! G \hookrightarrow R^k \tilde{f}_{t!} \tilde{j}_t^{-1} G \quad \text{for } t \geq t_o, \\ \text{(d)} \quad & \text{SS}_k(Rf_! G) \subset f_\pi \left(f_d^{-1} \left(\text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\ & \subset f_\pi f_d^{-1} (\text{SS}_k(G)). \end{aligned}$$

Proof. Let $\tilde{f}$ be the morphism $(f, \varphi): Y \rightarrow X \times \mathbf{R}$, and let q and $\tilde{\varphi}$ be the first and second projections defined on $X \times \mathbf{R}$ respectively. Then $\tilde{f}$ is proper on $\text{supp}(G)$ and $f = q \circ \tilde{f}$. This can be proved as follows. If K is a compact set of $X \times \mathbf{R}$, $\tilde{f}^{-1}(K)$ is compact.

Let i_t be the open embedding $X \times (-\infty, t) \hookrightarrow X \times \mathbf{R}$ and $\tilde{i}_t$ the closed embedding $X \times (-\infty, t] \hookrightarrow X \times \mathbf{R}$. Set $q_t = q \circ i_t$ and $\tilde{q}_t = q \circ \tilde{i}_t$. These morphisms can be visualized by the following diagrams.

$$\begin{array}{ccc} Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} \xrightarrow{q} X, \\ \uparrow j_t & & \uparrow i_t \nearrow q_t \\ \tilde{Y}_t & \xrightarrow{\tilde{f}_t} & X \times (-\infty, t) \end{array} \quad \begin{array}{ccc} Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} \xrightarrow{q} X. \\ \uparrow \tilde{j}_t & & \uparrow \tilde{i}_t \nearrow \tilde{q}_t \\ \tilde{Y}_t & \xrightarrow{\tilde{f}_t} & X \times (-\infty, t] \end{array}$$

Then, setting $\tilde{G} := Rf_*G$, we have

$$\begin{aligned} Rf_*G &\cong Rq_*\tilde{G}, & R\tilde{f}_*G &\cong R\tilde{q}_*\tilde{G}, \\ Rf_{t*}j_t^{-1}G &\cong Rq_{t*}i_t^{-1}\tilde{G}, & R\tilde{f}_{t*}\tilde{j}_t^{-1}G &\cong R\tilde{q}_{t*}\tilde{i}_t^{-1}\tilde{G}, \\ Rf_{t!}j_t^{-1}G &\cong Rq_{t!}i_t^{-1}\tilde{G}, & R\tilde{f}_{t*}\tilde{j}_t^!G &\cong R\tilde{q}_{t*}\tilde{i}_t^!\tilde{G}. \end{aligned}$$

Thus taking j -th cohomology, we have

$$\begin{aligned} R^j f_*G &\cong R^j q_*\tilde{G}, & R^j \tilde{f}_*G &\cong R^j \tilde{q}_*\tilde{G}, \\ R^j f_{t*}j_t^{-1}G &\cong R^j q_{t*}i_t^{-1}\tilde{G}, & R^j \tilde{f}_{t*}\tilde{j}_t^{-1}G &\cong R^j \tilde{q}_{t*}\tilde{i}_t^{-1}\tilde{G}, \\ R^j f_{t!}j_t^{-1}G &\cong R^j q_{t!}i_t^{-1}\tilde{G}, & R^j \tilde{f}_{t*}\tilde{j}_t^!G &\cong R^j \tilde{q}_{t*}\tilde{i}_t^!\tilde{G}. \end{aligned}$$

Therefore, it is enough to prove the result for $\tilde{G}$, q , i_t , $\tilde{\varphi}$ instead of G , f , j_t , φ .

(i) It follows from the hypothesis that

$$(6.1) \quad \text{SS}_k(\tilde{G}) \cap (T^*X \times \{(t; \tau); t > t_o\}) \subset T^*X \times \{(t; \tau); \tau \leq 0\}.$$

Indeed, by hypothesis, for any $(y; \eta) \in \text{SS}_k(G)$ and $(f(x); \xi) \in Y \times_X T^*X$, we have

$$(6.2) \quad d\varphi(y) \neq \eta + f^*\xi.$$

If $(x, t; \xi, \tau)$ is a point of $\text{SS}_k(\tilde{G}) \cap (T^*X \times \{(t; \tau); t > t_o\})$, then, by the formula (3.2), there exists $(y; \eta) \in \text{SS}_k(G)$ such that

$$\tilde{f}(y) = (x, t) \text{ and } \eta = \tilde{f}^*(\xi, \tau).$$

For this η , we have

$$\eta = \tilde{f}^*(\xi, \tau) = f^*\xi + \varphi^*\tau = f^*\xi + \tau d\varphi(y).$$

If $\tau > 0$, we get

$$d\varphi(y) = \frac{1}{\tau}\eta - \frac{1}{\tau}f^*\xi$$

and this contradicts (6.2) because $\text{SS}_k(G)$ is conic.

By (6.1), Let U be an open subset of X contained in a local chart, and convex in such a chart. By Proposition 5.1, we have isomorphisms for all $t > t_o$ and $j < k$

$$(6.3) \quad H^j(U \times \mathbf{R}; \tilde{G}) \xrightarrow{\sim} H^j(U \times (-\infty, t); \tilde{G})$$

and have an injection for all $t > t_0$.

$$(6.4) \quad H^k(U \times \mathbf{R}; \tilde{G}) \hookrightarrow H^k(U \times (-\infty, t); \tilde{G}).$$

This can be showed as follows.

This proves the result. □

Remark 6.2. $(U \times I_t)$ is not a gentle family.

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